

HOW THE SYSTEM WORKS (PROTOTYPE)

The Smart Hybrid Charcoal Cooler is a double-walled wooden crate with a ~10 cm gap filled with porous activated charcoal. The core working principle is evaporative cooling:

- A water reservoir (non-drip irrigation) slowly wets the charcoal bed.
- A 5V DC fan draws warm ambient air through the wet charcoal, causing evaporation and cooling the internal produce chamber by 10–15°C below ambient temperature.
- An SHT31 temperature and humidity sensor (I²C) monitors internal conditions in real time.
- An ESP32 DevKit C V4 microcontroller processes sensor data, dynamically controls the water pump (on/off) and fan speed to optimize cooling efficiency and battery life.
- A 0.96" OLED display (128×64, I²C) shows live readings (temperature, humidity, system status).
- A TP4056 battery charger/booster module manages a 3.7V LiPo battery, stepped up to 5V for the ESP32, with a small solar panel for off-grid recharging.
- A compact electrochemical ethylene (C₂H₄) sensor detects produce ripening gases.
- Alert thresholds trigger a 5V buzzer and red LED via GPIO17/GPIO19 when temperature or humidity exceed safe limits.
- All data is transmitted wirelessly and visualized on a real-time IoT dashboard (temperature trends, air quality, storage status, and predictive alerts), enabling quality certification documentation for market access.

This positions the system as an exceptionally low-cost alternative to conventional refrigeration, which is largely inaccessible to smallholder farmers.

Impact

Post-harvest losses account for 30–50% of fresh produce in Rwanda , costing farmers billions annually. The Smart Hybrid Charcoal Cooler directly addresses this by:

- Extending shelf life of vegetables and fruits by 3–7 days without grid electricity.
- Providing verifiable temperature/humidity logs via the IoT dashboard, enabling produce quality certification for premium market access.
- Reducing household food insecurity and increasing farmer income.
- Operating fully off-grid using solar power, making it viable in rural Rwanda and across the continent..

DASHBOARD INTEGRATION

The ESP32 transmits sensor data via Wi-Fi/MQTT to a cloud dashboard (web app) that provides:

- Live temperature and humidity inside the cooling systems
- Ethylene gas level monitoring for ripeness prediction.
- Automated alerts /SMS) when thresholds are breached.
- Downloadable quality certification reports for market compliance.
- Multi-unit fleet management for cooperatives managing many coolers.(after officially recognized by Rwanda

This dashboard layer is where significant long-term commercial value lies, and where your expertise in would be most impactful.

I have attached diagrams of the full prototype, including the physical system working diagram and the complete electronic schematic, for your reference.

I would be truly honored to discuss this further at your convenience whether a brief 20-minute call or a meeting I am confident that with you this prototype can evolve into a publishable research project and a commercially viable product with real-world development impact.

Thank you sincerely for your time and consideration.

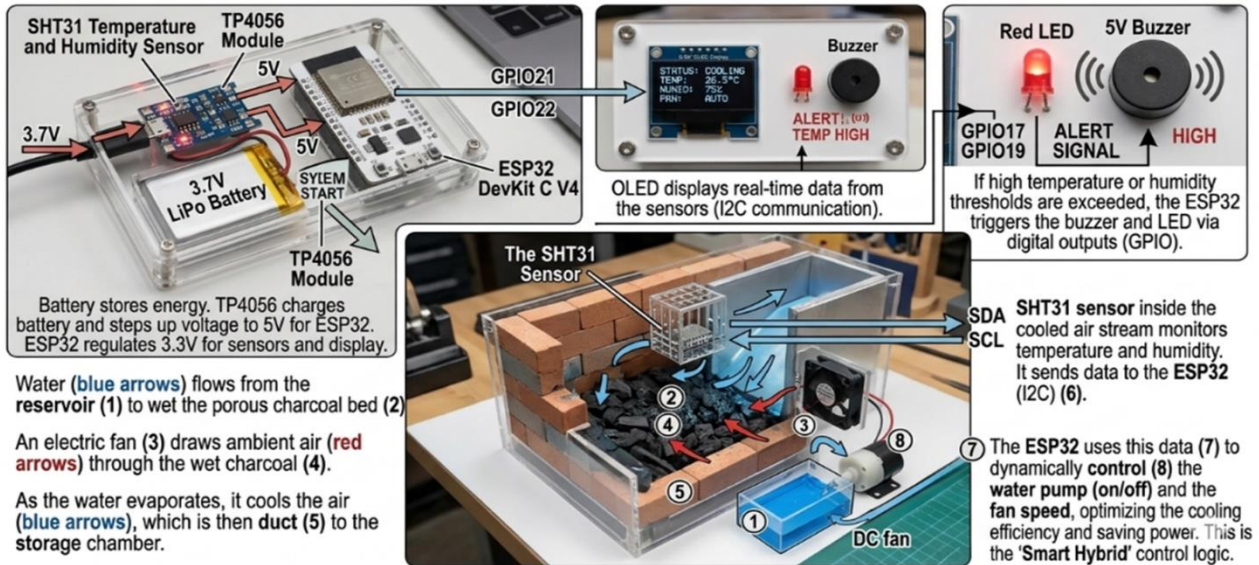
Warm regard

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Appendix



SMART HYBRID CHARCOAL COOLER: PHYSICAL SYSTEM WORKING DIAGRAM



SMART HYBRID CHARCOAL COOLER: FULL SCHEMATIC DIAGRAM

